

Yenho Chen

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Education

PhD Machine Learning, GEORGIA TECH, ATLANTA, GA

Aug. 2021 - May 2027 (Expected)

Advisor: Prof. Christopher J. Rozell

GPA: 4.0/4.0

Thesis: Dynamics-Based Self-Supervised Learning for Physiological Time Series

Recipient of University Presidential Fellowship

MS Computer Science, GEORGIA TECH, ATLANTA, GA

Aug. 2021 - Dec. 2023

Specialization in Visual Analytics

GPA: 4.0/4.0

BS Physics, UNIVERSITY OF TEXAS AT DALLAS, RICHARDSON, TX

Aug. 2015 - May 2019

Magna Cum Laude, Recipient of University Academic Excellence Award with Distinction

GPA: 3.9/4.0

Selected Papers

Y. Chen*, M. A. Xu, J. Regh, C. J. Rozell, "Self-Supervised Dynamical System Representations For Physiological Time-Series," *Accepted to the Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026.

Y. Chen*, B. Ancelin*, A. Falcon, P. Guan, C. Kaushik, N. Singh, B. Urcelay, "MANGO: Learning Disentangled Image Transformation Manifolds with Grouped Operators," *Sampling Theory and Applications (SampTA)*, 2025. (Oral)

Y. Chen*, N. Mudrik, A. S. Charles, C. J. Rozell, "Probabilistic Decomposed Linear Dynamical Systems for Robust Discovery of Latent Neural Dynamics," *In Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

Y. Chen*, N. Mudrik*, E. Yezerets, C. J. Rozell, A. S. Charles, "Decomposed Linear Dynamical Systems (dLDS) for learning the latent components of neural dynamics," *Journal of Machine Learning Research (JMLR)*, 2024.

* denotes first author or first co-authors

Work Experience

Research Assistant

2021 - Present

GEORGIA TECH

Atlanta, GA

- Led research on self-supervised learning methods for health time-series, using mathematical tools from dynamical systems theory to improve label efficiency and transferability of the learned representations across low-data and large-scale single- and multi-sensor settings.
- Initiated collaborations with Google, Rice University, and Johns Hopkins, producing first-author publications at **ICML**, **NeurIPS**, and **JMLR**.

Accelerated Discovery Research Intern

Summer 2025

IBM RESEARCH

San Jose, CA

- Designed a contrastive learning approach that uses evolutionary information from short linear motifs to predict phenotypes of mutated CAR-T sequences, enabling in silico screening for high-cytotoxicity variants and reducing de novo protein design costs.
- Submitted **a patent** and **a manuscript** by developing the core method in collaboration with experimentalists from Stanford and Altos Labs.

Artificial Intelligence Research Intern

Summer 2024

BLACKSTONE REAL ESTATE

New York City, NY

- Reduced Blackstone's hotel demand forecasting error by 26% with a novel differentiable dropout that contextually removes noisy modalities.
- Engineered a data ingestion pipeline with 35% faster throughput by optimizing SQL queries, accelerating ML experimentation.

Applied Large Language Model Intern

Spring 2024

TRAVELERS INSURANCE INC.

Atlanta, GA

- Achieved a 3x speedup in LLM pretraining efficiency by designing a domain reweighting method based on inter-domain gradient alignment.
- Increased evaluation throughput 5x by standardizing benchmarking across many HuggingFace LLM and reweighting method combinations.

Skills

Research

Self-supervised & contrastive learning, dynamical systems theory & state-space models, Bayesian inference, probabilistic graphical models, sparse coding, multimodal modeling, representation learning, transfer learning, semi-supervised learning

Applications

Health time-series (ECG, PPG, EEG, HAR, wearables), large language model (LLM) pretraining, protein language models

Software

PyTorch, Tensorflow, Scikit-Learn, Numpy, Python, C++, SQL, Linux, Git, Slurm, AWS, Databricks, Snowflake

Selected Research

Self-Supervised Dynamical Systems Pretraining for Multi-Sensor Health Time-Series

2026

MANUSCRIPT IN PREPARATION

Georgia Tech

- Developed a self-supervised pretraining framework for multi-sensor health time-series via a dynamics-based hybrid cross-reconstruction contrastive objective, with preliminary results showing 11% average error reduction in linear probe across 5 benchmark datasets.
- Derived theory on recovering transferable dynamical systems information from multi-sensor time-series datasets via a cross-reconstruction objective, motivating a novel observation process augmentation that isolates physiological dynamics from heterogeneous sensor configurations.

Self-Supervised Dynamical Systems Pretraining for Single-Sensor Health Time-Series

2025-2026

ACCEPTED AT ICML 2026

Georgia Tech

- Designed PULSE, a self-supervised framework for time-series that achieves a 12% error reduction across EEG, ECG, HAR, and PPG benchmarks in linear probe and semi-supervised settings, via a novel cross-reconstruction objective that isolates physiological dynamics from noise.
- Established the first theory relating cross-reconstruction with dynamical systems models of time-series to predict the efficacy of data augmentation strategies at recovering transferable information about the underlying physiological dynamics.

Sequence-Level CAR-T Phenotype Prediction via Soft-Contrastive Learning

2025-2026

SUBMITTED a Patent Application AND MANUSCRIPT TO CIBB 2026

IBM Research

- Extended CAR-T protein cytotoxicity prediction from motif- to sequence-level, achieving $R^2 = 0.705$ on held-out sequences, by designing a soft-contrastive objective that leverages BLASTp sequence similarities and consensus-region annotations.
- Outperformed ESM-2 by 32% in consensus-mutation classification, demonstrating higher sensitivity to biologically relevant point mutations.

Learning Disentangled Image Transformations with Grouped Manifold Operators

2024-2025

PUBLISHED AT SampTA 2025 (Oral)

Georgia Tech

- Developed MANGO, a manifold model that learns interpretable operators for controlling image transformations via a block-diagonal group structure, achieving a 60% improvement in out-of-distribution reconstruction over standard autoencoders.
- Reduced training time by 100x over the standard Manifold Autoencoder by replacing a 3-phase training procedure with a 1-phase approach.

Stable and Interpretable Neural Dynamics via Probabilistic dLDS

2023-2024

PUBLISHED AT NeurIPS 2024

Georgia Tech

- Developed a variational EM framework that estimates dLDS sparse codes robustly under dynamical noise in non-stationary systems, reducing the latent dynamics tracking error by 2x over the strongest baseline in simulation.
- Demonstrated discovery of emotion-aligned dynamic modes from LFP recordings that generalize to held-out data, where all prior methods fail.

Decomposing Linear Dynamical Systems (dLDS) for Interpreting Neural Dynamics

2022-2024

PUBLISHED AT JMLR 2024

Georgia Tech

- Introduced dLDS, an interpretable sparse dictionary learning framework over linear dynamical components supporting combinatorially more dynamic regimes than prior switching approaches for a fixed parameter budget.
- Demonstrated that dLDS disentangles independent subnetworks and identifies shared neural dynamics in *C. elegans* whole-brain recordings, solving a previously intractable task through the novel sparse operator formulation.

Selected Leadership

Bioengineering and Bioscience Unified Graduate Students (BBUGS)

2022 - Present

SOCIAL CHAIR

Georgia Tech

- Quadrupled event frequency (semesterly → monthly) for a 2K+ member community by directing a team of 5 and managing a \$6K annual budget.
- Established two annual traditions by reviving the overnight rafting trip post-COVID and co-founding the Garden Gala (100+ attendees each).

Reviewer Service

JOURNAL OF NEURAL ENGINEERING (2023), NEURIPS (2024,2025), ICML (2025, 2026), ICLR (2025, 2026)

Selected Awards

ICML GOLD REVIEWER AWARD

2026

NEURIPS TOP REVIEWER AWARD

2025

IBB "ABOVE AND BEYOND" TRAINEE AWARD

2025

GT MACHINE LEARNING STUDENT CONFERENCE (2ND PLACE POSTER)

2024

GEORGIA TECH PRESIDENT'S FELLOWSHIP

2021 - 2024

NIH ACADEMY SCHOLAR AWARD

2020